Accelerated Math II/III Name_____

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HW #5 - Polynomial Synthetic Division

Date_____ Period____

Divide using synthetic division. Write your answer in fraction form.

1) $(3x^5 - 6x^4 + 5x^3 + 8x^2 - 12x + 6) \div (x - 1)$

2) $(x^5 - 9x^4 + 16x^3 + 16x^2 - 25x + 17) \div (x - 6)$

3) $(8x^2 - 20x) \div (x - 1)$

4) $(x^2 + 3x - 20) \div (x - 4)$

5) $(11x^2 + 22x + 14) \div (x + 1)$

6) $(11x^3 - 21x^2 + x - 17) \div (x - 2)$

7) $(2x^5 + 14x^4 + 8x^3 - 27x^2 - 21x - 27) \div (x + 6)$

8) $(4x^5 - 21x^4 - 21x^3 + 13x^2 + 26x + 16) \div (x - 6)$

Determine if $d(x)$ is a factor of $f(x)$.

9) $f(x) = 2x^3 - 12x^2 + 5x + 28$
$d(x) = x - 5$

10) $f(x) = 4x^4 - 29x^3 + 29x^2 + 6x$
$d(x) = x - 6$

11) $f(x) = x^4 + 2x^3 - 11x^2 - 6x - 18$
$d(x) = x - 3$

12) $f(x) = 3x^3 - 2x^2 + 5x + 8$
$d(x) = x + 1$

Evaluate $f(x)$ at k.

13) $f(x) = 5x^4 - 6x^3 + 5x^2$
$k = 1$

14) $f(x) = x^4 - x^3 + 11x^2 - 11x$
$k = 2$

15) $f(x) = 6x^4 - 3x^2 - x + 2$
$k = 1$

16) $f(x) = 6x^3 + 26x^2 + 16x - 24$
$k = -1$

17) $f(x) = 3x^3 - 5x + 1$
$k = 1$

18) $f(x) = x^4 - 13x^3 + 28x^2 - 21x + 5$
$k = 2$

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Answers to HW #5 - Polynomial Synthetic Division

1) $3x^4 - 3x^3 + 2x^2 + 10x - 2 + \frac{4}{x-1}$

2) $x^4 - 3x^3 - 2x^2 + 4x - 1 + \frac{11}{x-6}$

3) $8x - 12 - \frac{12}{x-1}$

4) $x + 7 + \frac{8}{x-4}$

5) $11x + 11 + \frac{3}{x+1}$

6) $11x^2 + x + 3 - \frac{11}{x-2}$

7) $2x^4 + 2x^3 - 4x^2 - 3x - 3 - \frac{9}{x+6}$

8) $4x^4 + 3x^3 - 3x^2 - 5x - 4 - \frac{8}{x-6}$

9) No

10) Yes

11) Yes

12) No

13) 4

14) 30

15) 4

16) −20

17) −1

18) −13

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